

The Words of an Investigation

Circle Geometry · Grade 11 · what each word means, and how to write it so it earns the mark

Keep this one.

Every investigation uses these six words.

STEP 1

Observe

You measure. You notice.

STEP 2

Conjecture

You claim it happens always.

STEP 3

Prove

You argue it for every case.

STEP 4

Theorem

Now you may use it as a reason.

Underneath it all: axioms & assumed results. A proof leans on something earlier, which leans on something earlier again — and that chain has to stop somewhere. The facts we agree to accept without proving them are the floor of the whole building. In Grade 11 these are the things you may quote without proof: angles on a straight line add to 180° , vertically opposite angles are equal, the parallel-line angles, the congruency conditions.

Observation WHAT I SAW

A report of what actually happened in the cases you measured. It is honest and it is true — but it says **nothing at all** about the cases you did not try.

Sounds like: "When I dragged P and Q around the top of the circle, $\angle APB$ and $\angle AQB$ stayed equal."

Why it is not enough: it stops at the positions you happened to use.

Conjecture WHAT I CLAIM

A precise claim about **every** case, written so a stranger who has never seen your diagram could test it — so no P, no Q, no A, no B. Not yet proved, but now it is a real claim that can be attacked.

Sounds like: "Angles subtended by the same chord, at the circumference and on the same side of it, are always equal."

The marks live in the conditions, not in the answer.

Counterexample ONE CASE THAT KILLS IT

A single case where the conjecture fails. That is enough to destroy it for good, and nothing can repair it — because the claim said *always*, and now it doesn't.

Sounds like: "A line from the centre that bisects a chord is perpendicular to it" — **except the diameter**, which every line through the centre bisects. One case, conjecture dead.

Proof WHY IT MUST BE SO

An argument that covers **every** case at once, each step justified by a theorem you are allowed to quote. This is the only thing that can turn a conjecture into a theorem. Measuring never can.

The asymmetry: to *disprove*, find one case. To *prove*, handle all of them together. That is why the two jobs cost such different amounts of work.

Theorem PROVED, SO USABLE

A conjecture that has been proved. Only now may you write it as a **reason** in a rider — and only in the wording the marker accepts. Until it is proved it is not a reason, however many times you have measured it.

Accepted short forms: $\angle$ s in the same seg · $\angle$ at centre = $2 \times \angle$ at circumference · $\angle$ s in semi-circle · tans from same pt · opp $\angle$ s of cyclic quad

Converse THE SAME THEOREM, BACKWARDS

Swap what is **given** with what is **concluded**. That is a different claim travelling the other way, so proving the theorem proves nothing about its converse — it needs its own proof, and sometimes it is false.

Theorem: AB is a diameter $\rightarrow \angle ACB = 90^\circ$. **Converse:** $\angle ACB = 90^\circ \rightarrow$ AB is a diameter.

That is why every converse reason starts with the word *converse*.

Four learners saw the same thing. Only one wrote a conjecture.

- ✗ " $\angle APB = \angle AQB$."
An answer about one picture. It dies with the diagram.
- ✗ "When I dragged P and Q around, the angles stayed equal."
An observation. True — but it claims nothing beyond what was measured.
- ✗ "Any two angles standing on the same chord are equal."
A dropped condition. This quietly includes the angle at the **centre** — and that one is double, not equal.
- ✓ "Angles subtended by the same chord, at the circumference and on the same side of it, are **always equal**."
A conjecture: precise, general, and testable by a stranger.

The word that does the work

ALWAYS

Drop it and you have **reported a measurement**. Put it in and you have **made a claim**.

- It is what makes the sentence a conjecture at all.
- It is what makes it **disprovable** — one failing case now beats it.
- It is what makes a proof necessary, because "always" is a promise about cases you have never drawn.

"Numerous specific examples supporting a conjecture **do not constitute a general proof**."

IEB Mathematics SAGs — the one sentence this whole page exists to teach.